

SAI SNIGDHA YELLAPRAGADA

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A software developer committed to Agile practices with experience in writing clean and maintainable code using practices such as unit testing, continuous integration and deployment, automation, design patterns, refactoring, Test Driven Development (TDD) and a good understanding of object-oriented design principles such as patterns, cohesion/coupling, data structures and algorithms.

EDUCATION

- **Masters (MS) in Computer Science** at University of Alabama at Birmingham-Alabama, USA
 - **Bachelor of Technology in Computer Science** at GITAM Hyderabad-Telangana, INDIA
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EXPERIENCE

Software Engineer, SS Tech LLC.

Jan 29, 2024 - Present

- Involved in all phases of **Software Development Life Cycle** including requirements collection, design analysis of customer specifications and development of the application.
- Work with content management teams to author components and helped in replication process.
- Collaborate and work with other developers in teams to perform design, develop, code, deploy tests and debug.
- Involve in using core **Java** concepts- collection, exception handling, multi-threading, and serialization. Develop **back-end logic** using Java.
- Implemented **Postman** to define endpoints and call **RESTful web services**, effectively performing POST, PUT, DELETE, and GET methods for testing and debugging APIs.
- Develop Message Driven Bean components using **JMS** for e-mailing and Data Transfer between client and the providers.
- Manage **UAT** deployments in Jenkins and GIT.
- Maintain Git workflows for version control (**source code management**).
- Used **Splunk** to analyze application logs, identify error patterns, and create real-time monitoring dashboards for web services, helping to improve system performance and expedite troubleshooting.
- Leveraged **Kafka** for real-time data streaming and event-driven architecture, ensuring efficient communication between platform components and downstream systems.
- Develop, maintain, and automate **CI/CD pipelines** for code deployments using Jenkins in UAT and production.
- Used **JWT** for secure, stateless authorization, combined **OAuth 2.0** and **OpenID Connect** for safe authentication, and applied the latest security practices to protect data in microservices.
- Work in **DevOps/Agile** operations process and tools area (code review, unit test automation, service, incident and change management).
- Managing GitHub repositories and permissions including branching and tagging.
- Working with respective teams to resolve issues users are facing with **UAT** environments.
- Provision servers and deploy features using **Puppet/Chef/Ansible**.
- Built and deploy **Docker** containers for implementing micro-service architectures for Java applications.
- Successfully create and maintain automated CI/CD pipelines for code deployments using **Jenkins** while automating the deployment process resulting in the elimination of 70% of manual work and escalation of work efficiency.

- Participated in the planning, analysis, design, implementation, and maintenance of **AWS Fargate Agent** using **Go Language**.
- Worked in a sprint-based **Agile framework** and engaged in activities such as sprint planning, daily scrum, backlog refinement, sprint retrospective, and sprint review.
- Primary responsibilities include diving deep, designing, developing, and deploying medium to large sized projects.
- Other responsibilities include design reviews, writing clean, maintainable code, performing peer code-reviews, documenting, mentoring, optimizing performance, maintainability, and scalability.
- Proficient in **Go Language** and familiar with **AWS Infrastructure services and docker**.
- Managed and resolved production-related issues for AWS Fargate.
- Contributed to the development of **Linux OS**, focusing on **Kernel** and core components to enhance performance, security, and scalability for **Firecracker-based bare metal** environments.
- Performed debugging and optimization of boot processes, kernel drivers, networking, and storage stacks to improve system efficiency and reliability.
- Identified and implemented performance improvements and security hardening measures within the Linux OS and Firecracker instances.
- Conducted security assessments and applied mitigations to ensure a robust and secure operating environment.
- Was the main member in launching multiple prod regions within 4 months, which involved working with **cloud networking components** like VPCs, subnets, and route tables to configure regions for seamless service deployment and **tested end-to-end** (load testing, QA testing) and made them active regions.
- Worked extensively on **AWS IAM** (Identity and Access Management) to delegate required roles and permissions to AWS services.
- Worked with **AWS CDK** (Cloud development Kit) to implement AWS objects using **TypeScript**.
- Performed source code management and version control through **Git** and deployed changes in an automated **CI/CD** framework.
- During **on-call rotations**, debugged production issues by monitoring **AWS CloudWatch** alarms and tracing logs.
- Automated log gathering, service restarts, and issue triage using **Bash scripts**, and developed a **runbook** to help new recruits resolve similar issues efficiently, minimizing downtime and effort duplication.
- Created and maintained documentation for the projects to ensure that all team members were up to date on the latest developments.
- Demonstrated strong analytical skills by identifying root causes of complex technical issues and implementing efficient solutions.
- Developed new features for the Fargate Agent to meet changing business requirements.
- Adapted to evolving project requirements and changing priorities by quickly adjusting development strategies. Effectively managed shifting timelines and scope modifications, ensuring timely delivery of projects without compromising quality.
- Monitored performance of the Fargate Agent to ensure that it met user expectations.
- Deployed updates in a timely manner to ensure that users had access to the latest features.
- Resolved business systems issues by working closely with stakeholders to understand their needs.
- Projects taken up include **Region Builds, Enhancement of Functional Tests, and Fargate Agent Feature Enhancement**.

Teaching Assistantship, University of Alabama at Birmingham

Jul 2021 – May 2022

- Worked as Teaching Assistant for **Cloud Security** in Fall 2021 and **Networking** in Spring 2022 in the Computer Science Department under the faculty supervision.
- Responsibilities: Tutoring, handling lab/hands-on sessions, grading tests, and assisting teaching staff with practical demonstrations.
- Developing/finalizing homework assignments and examinations.
- Technologies used are **AWS cloud services**, **IoT (Cloud Security)**, **Cisco Packet Tracer (Networking)**.
- Used **Java** and **Python** as the programming languages and **Git** for continuous deployment.
- Have hands on experience in stored procedures on production **PostgreSQL** that results in quicker results and better understanding of the working of networking devices.

Internship at Infochord Technologies Pvt. Ltd.

May 2019 – Nov 2019

- Worked as a Full Stack Web Developer.
- The task was to development a CRUD application, Online Blogging Application in this case.
- Used **Bootstrap**, **JSON**, **HTML**, and **CSS** for front-end development, **PHP**, **JavaScript**, and **REST API** for back-end development, and **MySQL** for database creation and maintenance.
- Used **AJAX** for asynchronous execution of HTTP requests from client-side.

TECHNICAL SKILLSET

Programming Languages	:	C, Java, Python, Golang, basics of C++, Swift, Ruby
Software Framework	:	Bootstrap, Flask, Hadoop
Database & Servers	:	SQLite, MySQL, PostgreSQL, XAMPP
Web Development	:	HTML, CSS, XML, PHP, JavaScript, TypeScript, API
Tools & Technologies	:	Salesforce, DevOps, IntelliJ, GitHub, AWS, Docker

CERTIFICATIONS & ACHIEVEMENTS

- Awarded top scorer of test conducted by FACE
- Received Best Student award in school
- Certifications in Python and SQLite from University of Michigan and Coursera
- Certification in Artificial Intelligence from Automation Anywhere
- Certification in Psychology from Yale University and Coursera
- Certification in Network Virtualization concepts and Software Defined Storage Concepts from VMWare
- Certification in Cyber Security Essentials from Cisco Networking Academy
- Certification in Entrepreneurship from Cisco Networking Academy
- Certification in Deep Learning and Machine Learning from MathWorks
- Mountaineer rank in Salesforce trailheads

EXTRA CURRICULARS

- Member of The University of Alabama at Birmingham's Off Campus Student Council
- Member of National Social Scheme, India
- Volunteered as the web casting officer for the 2019 general elections in India